

The Discrete-Time Fourier Transform

1 Representation of DTFT

Definition 1.1 (The Fourier Transform Pair of a DT Aperiodic Signal).

If an aperiodic signal $x[n]$ has a Fourier transform, then

$$x[n] = \frac{1}{2\pi} \int_{2\pi} X(e^{j\omega}) e^{j\omega n} d\omega \quad (\text{The Synthesis Equation})$$

$$X(e^{j\omega}) = \sum_{n=-\infty}^{+\infty} x[n] e^{-j\omega n} \quad (\text{The Analysis Equation})$$

where the function $X(e^{j\omega})$ is referred to as the *discrete-time Fourier transform* or the *spectrum* of $x[n]$.

Since $e^{j\omega}$ has a period of 2π , $X(e^{j\omega})$ is always periodic with a period 2π . $\omega = 0$ and $\omega = 2\pi$ yield the same signal. Therefore, usually we only consider the spectrum from $\omega = -\pi$ to $\omega = \pi$, and signals at frequencies near $\omega = 0$ or $\pm 2\pi$ can be considered as **low-frequency**, while signals at frequencies near $\omega = \pm\pi$ can be considered as **high-frequency**.

2 Fourier Transform for Periodic Sequences

Theorem 2.1.

For a periodic sequence $x[n]$ whose FS coefficients are $\{a_k\}$, the Fourier transform $X(e^{j\omega})$ is given by

$$X(e^{j\omega}) = \sum_{k=-\infty}^{\infty} 2\pi a_k \delta(\omega - k\omega_0)$$

Proof.

$$\begin{aligned} X(e^{j\omega}) &\xrightarrow{\mathcal{F}^{-1}} \frac{1}{2\pi} \int_{2\pi} X(e^{j\omega}) e^{j\omega n} d\omega = \frac{1}{2\pi} \int_W^{W+2\pi} \sum_{k=-\infty}^{\infty} 2\pi a_k \delta(\omega - k\omega_0) e^{j\omega n} d\omega \\ &= \sum_{k=-\infty}^{\infty} a_k \int_W^{W+2\pi} \delta(\omega - k\frac{2\pi}{N}) e^{jk\omega_0 n} d\omega \\ &= \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 n} \left(\begin{cases} 1 & \text{for } W \leq k\frac{2\pi}{N} < W+2\pi \\ 0 & \text{otherwise} \end{cases} \right) \\ &= \sum_{k=\langle N \rangle} a_k e^{jk\omega_0 n} \\ &= x[n] \end{aligned}$$

□

3 Properties of CT Fourier Transform

3.1

Suppose we have:

$$\begin{cases} x[n] \xleftrightarrow[\mathcal{F}^{-1}]{\mathcal{F}} X(e^{j\omega}) \\ y[n] \xleftrightarrow[\mathcal{F}^{-1}]{\mathcal{F}} Y(e^{j\omega}) \end{cases}$$

Then we have the following results:

Linearity

$$a \cdot x[n] + b \cdot y[n] \xleftrightarrow[\mathcal{F}^{-1}]{\mathcal{F}} a \cdot X(e^{j\omega}) + b \cdot Y(e^{j\omega})$$

Time Shifting

$$x[n - n_0] \xleftrightarrow[\mathcal{F}^{-1}]{\mathcal{F}} e^{-j\omega n_0} X(e^{j\omega})$$

Frequency Shifting

$$e^{j\omega_0 n} x[n] \xleftrightarrow[\mathcal{F}^{-1}]{\mathcal{F}} X(e^{j(\omega - \omega_0)})$$

Time Reversal

$$x[-n] \xleftrightarrow[\mathcal{F}^{-1}]{\mathcal{F}} X(e^{-j\omega})$$

Scaling

$$x_{(k)}[n] = \begin{cases} x[\frac{n}{k}] & \text{for } \frac{n}{k} \in \mathbb{Z} \\ 0 & \text{otherwise} \end{cases} \xleftrightarrow[\mathcal{F}^{-1}]{\mathcal{F}} X(e^{jk\omega})$$

Conjugation

$$x^*[n] \xleftrightarrow[\mathcal{F}^{-1}]{\mathcal{F}} X^*(e^{-j\omega})$$

Convolution

$$x[n] * y[n] \xleftrightarrow[\mathcal{F}^{-1}]{\mathcal{F}} X(e^{j\omega}) Y(e^{j\omega})$$

Multiplication

$$x[n] y[n] \xleftrightarrow[\mathcal{F}^{-1}]{\mathcal{F}} \frac{1}{2\pi} \int_{2\pi} X(e^{j\theta}) Y(e^{j\omega - j\theta}) d\theta$$

Differencing in Time

$$x[n] - x[n - 1] \xleftrightarrow[\mathcal{F}^{-1}]{\mathcal{F}} (1 - e^{-j\omega}) X(e^{j\omega})$$

Accumulation in Time

$$\sum_{k=-\infty}^n x[k] = x[n] * u[n] \xleftrightarrow[\mathcal{F}^{-1}]{\mathcal{F}} \frac{1}{1 - e^{-j\omega}} X(e^{j\omega}) + \pi X(e^{j0}) \sum_{k=-\infty}^{\infty} \delta[\omega - 2\pi k]$$

Differentiation in Frequency

$$n \cdot x[n] \xleftrightarrow[\mathcal{F}^{-1}]{\mathcal{F}} j \frac{d}{d\omega} X(e^{j\omega})$$

Even-Odd Decomposition for Real Signals

$$x[n] \text{ real: } \mathcal{E}v\{x[n]\} \xleftrightarrow[\mathcal{F}^{-1}]{\mathcal{F}} \mathcal{R}e\{X(e^{j\omega})\}$$

$$x[n] \text{ real: } \mathcal{O}d\{x[n]\} \xleftrightarrow[\mathcal{F}^{-1}]{\mathcal{F}} j \mathcal{I}m\{X(e^{j\omega})\}$$

Parseval's Relation for Aperiodic Signals

$$\int_{-\infty}^{\infty} |x[n]|^2 dn = \frac{1}{2\pi} \int_{2\pi} |X(e^{j\omega})|^2 d\omega$$

3.2 Symmetry

- $x[n] \text{ real} \iff X(e^{j\omega}) = X^*(e^{-j\omega})$
- $x[n] \text{ purely imaginary} \iff X(e^{j\omega}) = -X^*(e^{-j\omega})$

Conjugate Symmetry For real $x[n]$, its FT $X(e^{j\omega})$ satisfies

- $X(e^{j\omega}) = X^*(e^{-j\omega})$
- $\mathcal{Re}\{X(e^{j\omega})\} = \mathcal{Re}\{X(e^{-j\omega})\}$
- $\mathcal{Im}\{X(e^{j\omega})\} = -\mathcal{Im}\{X(e^{-j\omega})\}$
- $|X(e^{j\omega})| = |X(e^{-j\omega})|$
- $\arg X(e^{j\omega}) = -\arg X(e^{-j\omega})$

Parity

- $x[n]$ even $\iff X(e^{j\omega})$ even
- $x[n]$ odd $\iff X(e^{j\omega})$ odd
- $x[n]$ real & even $\implies X(e^{j\omega})$ real & even
- $x[n]$ real & odd $\implies X(e^{j\omega})$ purely imaginary & odd

4 Basic Fourier Transform Pairs

$$\text{sinc}(\theta) := \frac{\sin(\pi\theta)}{\pi\theta}$$

4.1 Table

Signal	Fourier Transform	Fourier Series Coefficients
$\sum_{k \in \langle N \rangle} a_k e^{jk\omega_0 n}$	$2\pi \sum_{k=-\infty}^{\infty} a_k \delta(\omega - k\omega_0)$	a_k
$e^{j\omega_0 n}$	$2\pi \sum_{l=-\infty}^{\infty} \delta(\omega - \omega_0 - 2\pi l)$	if $\frac{\omega_0}{2\pi} = \frac{m}{N}$, $a_k = \begin{cases} 1 & k = m, m \pm N, m \pm 2N, \dots \\ 0 & \text{otherwise} \end{cases}$ if $\frac{\omega_0}{2\pi} \notin \mathbb{Q}$, a_k is aperiodic
$\cos(\omega_0 n)$	$\pi \sum_{l=-\infty}^{\infty} \delta(\omega - \omega_0 - 2\pi l) + \delta(\omega + \omega_0 - 2\pi l)$	
$\sin(\omega_0 n)$	$\frac{\pi}{j} \sum_{l=-\infty}^{\infty} \delta(\omega - \omega_0 - 2\pi l) - \delta(\omega + \omega_0 - 2\pi l)$	
Periodic Square Wave $\begin{cases} 1 & n \leq N_1 \\ 0 & N_1 < n \leq N/2 \end{cases}$ with period of N	$2\pi \sum_{k=-\infty}^{\infty} a_k \delta(\omega - k\omega_0)$	$a_k = \begin{cases} \frac{(2N_1 + 1) \sin(k\omega_0)}{2N \sin(k\omega_0/2)} & \text{otherwise} \\ \frac{2N_1 + 1}{N} & k = 0, \pm N, \pm 2N, \dots \end{cases}$
$\sum_{k=-\infty}^{+\infty} \delta[n - kN]$	$\frac{2\pi}{N} \sum_{k=-\infty}^{+\infty} \delta\left(\omega - \frac{2\pi k}{N}\right)$	$a_k = \frac{1}{N}$
$\begin{cases} 1 & n \leq N_1 \\ 0 & N_1 < n \end{cases}$	$\frac{\sin[\omega(N_1 + 1/2)]}{\sin(\omega/2)}$	
$\frac{\sin Wn}{\pi n}$ $0 < W < \pi$	$\begin{cases} 1 & \omega \leq W \\ 0 & W < \omega \leq \pi \end{cases}$ with period of 2π	
1	$2\pi \sum_{l=-\infty}^{\infty} \delta(\omega - 2\pi l)$	$a_k = \begin{cases} 1 & k = 0, \pm N, \pm 2N, \dots \\ 0 & \text{otherwise} \end{cases}$
$\delta[n]$	1	
$u[n]$	$\frac{1}{1 - e^{-j\omega}} + \pi \sum_{l=-\infty}^{\infty} \delta(\omega - 2\pi l)$	
$\delta[n - n_0]$	$e^{-j\omega n_0}$	
$a^{ n }$ with $ a < 1$	$\frac{1}{1 - ae^{-j\omega}} + \frac{1}{1 - ae^{j\omega}} - 1$	
$a^n u[n]$ with $ a < 1$	$\frac{1}{1 - ae^{-j\omega}}$	
$(n+1)a^n u[n]$ with $ a < 1$	$\frac{1}{(1 - ae^{-j\omega})^2}$	
$\binom{n+r-1}{n} a^n u[n]$ with $ a < 1$	$\frac{1}{(1 - ae^{-j\omega})^r}$	

4.2 Some Useful Techniques

$$\delta[n] = \frac{\sin(\pi n)}{\pi n} = \text{sinc}(n)$$

Expand the delta function to complex domain:

$$\delta(z) := \delta(\text{Re}\{z\})\delta(\text{Im}\{z\}) \quad \text{for } z \in \mathbb{C}$$

Then we have:

$$\sum_{l=-\infty}^{\infty} \delta(\omega - 2\pi l) = \delta(e^{j\omega} - 1) \quad \sum_{l=-\infty}^{\infty} \delta(\omega + \omega_0 - 2\pi l) = \delta(e^{j(\omega+\omega_0)} - 1) \quad \sum_{k=-\infty}^{+\infty} \delta\left(\omega - \frac{2\pi k}{N}\right) = \delta(e^{jN\omega} - 1)$$

$$\begin{aligned} e^{j\omega_0 n} & \xleftrightarrow[\mathcal{F}^{-1}]{\mathcal{F}} 2\pi\delta(e^{j(\omega-\omega_0)} - 1) \\ \cos(\omega_0 n) & \xleftrightarrow[\mathcal{F}^{-1}]{\mathcal{F}} \pi \left[\delta(e^{j(\omega-\omega_0)} - 1) + \delta(e^{j(\omega+\omega_0)} - 1) \right] \\ \sin(\omega_0 n) & \xleftrightarrow[\mathcal{F}^{-1}]{\mathcal{F}} \frac{\pi}{j} \left[\delta(e^{j(\omega-\omega_0)} - 1) - \delta(e^{j(\omega+\omega_0)} - 1) \right] \\ \sum_{k=-\infty}^{+\infty} \delta[n - kN] & \xleftrightarrow[\mathcal{F}^{-1}]{\mathcal{F}} \frac{2\pi}{N} \delta(e^{jN\omega} - 1) \\ 1 & \xleftrightarrow[\mathcal{F}^{-1}]{\mathcal{F}} 2\pi\delta(e^{j\omega} - 1) \\ u[n] & \xleftrightarrow[\mathcal{F}^{-1}]{\mathcal{F}} \frac{1}{1 - e^{-j\omega}} + \pi\delta(e^{j\omega} - 1) \end{aligned}$$

$$a_k = \frac{X(e^{jk\omega_0})}{2\pi\delta(e^{j2\pi k} - 1)} \xleftrightarrow[\mathcal{FS}]{\mathcal{FS}^{-1}} x[n] \xleftrightarrow[\mathcal{F}^{-1}]{\mathcal{F}} X(e^{j\omega}) = 2\pi a_{\omega/\omega_0} \delta(e^{jN\omega} - 1)$$

5 Systems Characterized By LCCDEs

An LTI system with initial rest characterized by the LCCDE

$$\sum_{k=0}^N a_k y[n - k] = \sum_{k=0}^M b_k x[n - k]$$

can be represented in the frequency domain as

$$H(e^{j\omega}) = \frac{Y(e^{j\omega})}{X(e^{j\omega})} = \frac{\sum_{k=0}^M b_k e^{-j\omega k}}{\sum_{k=0}^N a_k e^{-j\omega k}}$$